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DIANE E. BAILEY

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Modeling work group effectiveness in high-technology manufacturing environments

DIANE E. BAILEY

Department of Industrial Engineering and Engineering Management, Stanford University, Stanford, CA 94305-4024, USA
E-mail: Diane.Bailey@stanford.edu

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Existing models of work group effectiveness have been tested in low-technology manufacturing settings, but not in a high-technology one. Typical features of high-technology environments – substantial automation, computerized scheduling systems, complex production processes, capital-intensive production, high-cost equipment, and high product value – render them significantly different from their low-technology counterparts. The increased use of work groups and teams among high-technology manufacturers raises the question of whether existing models can be generalized to these settings. In this paper, data from 89 groups in the semiconductor manufacturing industry are used to create predictive models of group productivity, job satisfaction, and perceived performance. External variables (such as conflict with supervisors and engineers) are found to be more predictive of productivity, while internal variables (such as conflict among group members) are more predictive of satisfaction. These results highlight the importance of fostering the work group's interaction with external technical support personnel when designing and managing successful work groups in high-technology workplaces.

1. Introduction

In high-technology manufacturing environments, the tasks of shop-floor operators are evolving [1]. Traditionally performing primarily equipment operation tasks, operators are now frequently involved in monitoring, analytical problem-solving, and trouble-shooting assignments. These changes are occurring for a number of reasons. The implementation of lean manufacturing practices, such as Just-In-Time (JIT) inventory policies, and the development of integrated automated production and information systems supplant the worker's control over the pace of work [2]. Workers no longer decide what to produce next; rather, systems make these decisions for them. At the same time, the recent emphasis on quality and continuous improvement requires operators to expand their skill sets so that they might analyze and solve recurring problems in production. Thus, rather than being de-skilled by manufacturing advances, operators are finding their jobs enlarged by them. Kern and Schumann [3] call workers in such transformed jobs "systems controllers," and report their emerging presence in electronics, automobile, chemical, and machine manufacturing industries.

Under most quality paradigms, workers are trained in areas such as statistical process control, JIT inventory policies, continuous improvement, and preventive maintenance [4]. Operator involvement in these new manu-

facturing practices often alters workplace dynamics among operators, supervisors, engineers, technicians, and others on the shop floor. For example, operators may be given greater technical autonomy as the result of new training, thereby shifting decision-making control away from supervisors and engineers. Often this new authority is not granted to individuals *per se* but rather to the production work group as a unit.

To date, most of the empirical literature on production work groups has focused on low-technology industries (e.g., a confectionery factory [5], a locomotive shop [6]). In such industries, the shop-floor ratio of engineers and technicians to operators is typically quite low. Processes are stable, technological innovations are infrequent, and production can be overseen by production supervisors with limited engineering intervention. An apparel factory is a good example. There, one might find one engineer on a shop floor with 130 operators and several supervisors. Changes in equipment and sewing technology are rare; productivity improvements focus on reducing the sewing time at each operation via traditional work and time study methods. Product value and capital investment are low. Automation is made difficult due to shifting in the cutting of flexible fabrics, resulting in non-uniform pieces that operators must tug into position while sewing. Constant changes in fashion styles may cause scheduling and delivery problems at the factory, but sewing processes themselves change very little.

By contrast, high-technology manufacturing workplaces exhibit increasing trends towards automated and highly complex production processes, high product value, large capital investment, rapid changes in product mix, short learning curves, intense competition, and a high proportion of technically-skilled employees among the production workforce. In the semiconductor industry, for example, a production facility with 130 operators may have a staff of 30 engineers in addition to a dozen supervisors. Equipment may cost a million dollars or more per machine. Technology advances rapidly, leading to frequent changes in both product and processes. To prevent costly human error, equipment is linked to remove set-ups, products are bar-coded to permit electronic downloading of process instructions, and computerized scheduling systems may dictate job scheduling at each machine. Improvement efforts focus on reducing cycle time, costs, and scrap; they often involve shop-floor operators in improvement programs that feature continuous improvement teams, autonomous work groups, quality teams, or other types of work groups [7].

In examining what makes work groups effective, the literature traditionally has focused on internal group dynamics, such as the development of group norms, conflict among group members, and the emergence of group cohesion [8]. Yet, Hackman [9] stresses that production groups are unique among groups in that, while they have an internal orientation, their development is spurred by external interactions. Other researchers have called for greater investigation of these interactions, particularly those that occur along the margins of a group's boundaries [10,11]. One of the few external relationships that has received much attention is that between the work group and the production supervisor. In the wake of increasing team autonomy, supervisors have been transformed from bosses who give orders to facilitators who guide team actions [12]. Very little attention has been paid to other external relationships, such as those between the work group and the engineers with whom it interfaces. Conceivably, this absence in the literature is a direct result of the concentration of research in low-technology environments, where the role of the engineering staff is much lower. In high-technology industries, a group's interaction with engineers, technicians, and other technical support staff might have significant ramifications on its performance.

In this paper, the predictors of production work group performance in high-technology work environments are examined. Particular attention is paid to external relationships by separating common variables, such as communication and feedback, into internal and external components. A model of group performance is presented and tested with data from 89 production teams in the semiconductor industry. The paper concludes with implications for industry and ideas for future research.

2. Model description

The model presented in Fig. 1 extends the input-process-output tradition established by McGrath [13] via the addition of the bottom arc linking design factors with dependent outcomes. Thus, design factors are thought to influence outcomes directly, as well as through the mediation of group processes.

To capture the external interactions that are suspected to have great import for high-technology work groups, this model reinterprets several traditional variables regarding work group design and group processes. This reinterpretation is accomplished via the separation of these constructs into external and internal components. For example, task dependence (how much an individual team member relies on someone other than himself to complete his work) is separated into an external component for dependence on agents outside the production work group (e.g., equipment technicians, production supervisors, and engineers) and an internal component for dependence on members strictly within the group. Performance feedback is similarly separated along external-internal lines, as are communication, conflict, and supportiveness of the employee involvement team program.

Several other variables have been subdivided not to capture external relationships, but to reflect differences between task-centered and group-centered influences. Thus, training is divided into task training, equipment cross training, and team training (instruction in topics such as conflict resolution). Group autonomy is separated into technical and administrative components, to reflect differences between making decisions over the product, process, or equipment versus making decisions regarding group discipline and other personnel issues. Cohesion is separated into an affective component to capture how much group members like one another, and a task-oriented component to measure their mutual commitment to the group's goal. These separations are made so that we might examine how strong the influence of technical or task-based factors are for high-technology work groups.

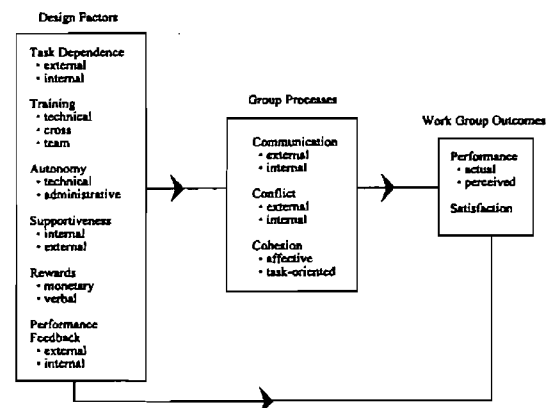


Fig. 1. Model of work group effectiveness.

Two variables are separated into components that are intended to provide distinctions between extrinsic and intrinsic elements. Reward valence (the belief that good performance will lead to rewards) is separated into monetary valence (the belief that good performance will lead to an increase in pay or bonuses) and verbal valence (the belief that good performance will lead to praise and recognition). Similarly, satisfaction's extrinsic component measures happiness with pay, while its intrinsic component measures all other forms of happiness in the job, such as contentment with supervision and opportunities for growth. These separations will allow us to examine the importance of intrinsic elements for shop-floor workers in high-technology environments.

Finally, performance is measured both as actual and perceived performance. Actual performance is that which can be objectively measured, such as the number of goods produced in a given time. Perceptions of performance are subjective measurements that may be garnered from the work group, from management, or from customers.

In addition to testing the overall model, this paper will also test the following hypothesis:

External variables (e.g., external communication, external conflict, external task dependence) will prove important predictors of performance and attitudes for high-technology work groups.

3. Methodology

3.1. Sample

The level of analysis in this study is the work group. The sample is drawn from eight semiconductor fabrication facilities (fabs) representing seven major firms. Three of the eight fabs were selected through their participation in a large benchmarking study of the industry [14], the other five were identified through their membership in industry associations. Similar process technologies, products, and chip line widths ensured a nearly identical equipment set among the sample, with production bottlenecks appearing in the same areas. This similarity, in addition to full capacity production at each site, facilitated quantitative comparisons. A pilot fab served as a site to become familiar with work organization and production aspects of semiconductor manufacturing. A written survey (described below) was administered to operators at the pilot site and modified accordingly.

3.2. Data collection

Data for the study were gathered during a 4- to 5-day site visit made to each fab during the period 1992–1993. Quantitative performance data covering 1 to 3 months, depending on availability, were gathered from production records. Qualitative data were additionally gathered as a

check on the quantitative analysis. These data were gained in part via semi-structured hour-long interviews with employees in every job type within the engineering, production, and equipment hierarchies at each fab. Over 100 interviews were held in total. At least three team meetings were also attended and observed during each visit, and 4-hour unguided tours of the manufacturing areas were taken to permit informal discussions with operators. Finally, employee surveys were distributed to two to four operators per work group at each fab. Multiple responses were averaged together to serve as the group response.

3.2.1. Production records and workforce data

Direct labor productivity was measured as the number of wafers processed per operator-hour. Indirect labor productivity was calculated as the number of wafers processed per operator, supervisor, and technician-hour combined.

3.2.2. Self-report measures

Self-report measures were gathered for the design factors and group process variables, as well as for three of the outcome variables. Survey items were measured on 5-point Likert-type scales [15]. An example is given in Fig. 2; although the anchors of "quite" versus "very" supportive may appear indistinct, responses suggest that operators made distinctions between the two levels. The questions in the survey were modified from van de Ven and Ferry's [16] instrument for measuring organizational effectiveness at the work group level. Two variables not included in the model, education and organizational tenure, were included as control variables and measured via the survey.

3.2.3. Response rates

Eighty-nine work groups (72%) submitted self-report questionnaire data in addition to their production and interview data. On average, just over one-quarter of the operator workforce was surveyed at each site.

3.3. Analysis

Pearson correlation coefficients (i.e., the covariance of two variables expressed in z scores, hereafter simply

1. How supportive have the following people been of team activities involving operators in your fab?					
	NOT AT ALL SUPPORTIVE	A LITTLE SUPPORTIVE	SOMEWHAT SUPPORTIVE	QUITE SUPPORTIVE	VERY SUPPORTIVE
a. Supervisors	1	2	3	4	5
b. Technicians	1	2	3	4	5
c. Engineers	1	2	3	4	5
d. Your fellow operators	1	2	3	4	5

Fig. 2. Sample question from the survey.

correlations) served as a check of the direct link between design factors and outcomes. A separate regression model was then built for each of the five outcome variables. Backwards regression with listwise deletion of missing cases was performed in each instance.

4. Results

4.1. Correlation

Correlations, means, and standard deviations are given in Table 1. Correlations with outcome variables that are significant at the 0.05 level or below are shown in bold. In looking along the outcome variable rows, we see that the direct link between design factors and work group outcomes is supported for perceived performance and intrinsic satisfaction, as indicated by the large number of significant correlations between design factors and these variables. Technical and administrative autonomy, external feedback, supportiveness, both internal and external, and team training are positively correlated with both perceived performance and intrinsic satisfaction.

However, the direct link with design factors is not supported for labor productivity (both direct and indirect) and extrinsic job satisfaction. For these outcomes, we see only a few significant correlations with design factors.

4.2. Regression models

Regression model results are presented in Table 2. The significance levels of the *F*-statistics are all well below the 0.05 level, indicating that the models reflect a good fit of the data. Not shown are residual plots (of each variable in the equations against the respective dependent variable) and normal probability plots that indicate that the models do not violate assumptions of linearity and normality and equality of variance. Collinearity diagnostics (i.e., checks for multicollinearity, or high correlations among independent variables) indicate low multiple correlation among variables both included in and dropped from the final regression models. The model for intrinsic job satisfaction, with seven variables remaining in the equation, shows the highest adjusted *R* squared value at 0.55.

Table 1. Correlations, means, and standard deviations

	<i>Mean</i>	<i>Standard deviation</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>
<i>Predictors</i>									
1 Autonomy-administrative	2.76	0.85	1.00						
2 Autonomy-technical	2.46	0.70	0.73	1.00					
3 Cohesion-affective	2.88	0.58	0.29	0.31	1.00				
4 Cohesion-task	2.63	0.48	0.35	0.38	0.26	1.00			
5 Communication-external	2.79	0.66	0.18	-0.05	0.15	0.18	1.00		
6 Communication-internal	3.18	0.83	0.37	0.22	0.21	0.18	0.58	1.00	
7 Conflict-external	1.82	0.69	-0.21	-0.13	-0.15	0.05	0.08	0.13	1.00
8 Conflict-internal	2.06	0.90	0.01	0.02	-0.05	0.07	-0.10	0.16	0.60
9 Education	2.54	0.66	0.10	0.07	0.06	-0.04	0.14	0.26	0.03
10 Feedback-external	3.02	0.76	0.00	0.07	0.13	0.20	0.22	0.18	0.14
11 Feedback-internal	2.27	0.98	0.26	0.35	0.29	0.29	0.19	0.33	0.17
12 Organizational tenure	3.66	0.98	-0.11	0.06	0.06	0.17	-0.14	-0.06	-0.08
13 Reward-monetary valence	2.55	0.84	-0.22	-0.17	-0.02	0.00	0.05	0.06	0.04
14 Reward-verbal valence	2.84	0.79	0.03	0.04	-0.02	0.32	0.13	-0.06	-0.06
15 Supportiveness-internal	3.57	0.96	0.55	0.55	0.30	0.33	0.17	0.22	-0.15
16 Supportiveness-external	2.64	0.84	0.46	0.46	0.32	0.49	0.07	0.21	-0.19
17 Task dependence-external	1.92	0.40	0.25	0.26	-0.07	0.15	0.30	0.24	0.20
18 Task dependence-internal	2.37	0.65	0.41	0.43	0.05	0.08	0.14	0.39	0.15
19 Training-cross	3.43	0.83	0.45	0.51	0.06	0.37	0.07	0.15	-0.16
20 Training-team	2.48	1.41	0.48	0.45	0.09	0.10	-0.02	0.22	-0.10
21 Training-task	3.48	0.93	0.05	0.03	0.17	0.13	0.14	-0.07	-0.03
<i>Outcomes</i>									
22 Labor productivity-direct	33.47	17.26	-0.15	-0.16	-0.02	-0.06	-0.18	-0.18	0.15
23 Labor productivity-indirect	24.02	12.16	-0.02	-0.01	-0.01	0.04	-0.13	-0.05	0.14
24 Perceived performance	3.63	0.60	0.28	0.39	0.28	0.24	0.06	0.12	-0.13
25 Job satisfaction-extrinsic	2.77	0.80	0.14	0.11	0.15	0.07	0.22	0.13	-0.10
26 Job satisfaction-intrinsic	3.56	0.53	0.32	0.26	0.21	0.18	0.23	0.17	-0.22

4.3. Hypothesis

In terms of the hypothesis, which contends that external variables will prove key predictors of performance and attitudes, we find that external variables do not predict either intrinsic or extrinsic job satisfaction. However, they do appear in the models for actual and perceived performance. External conflict appears in both productivity models, while external feedback and external supportiveness appear in the model for indirect labor productivity. The model for perceived performance features external communication and external task dependence.

In contrast to external variables, internal variables dominate the models for job satisfaction. In fact, there seems to be a rather clear separation: external variables predict productivity, internal ones predict satisfaction. Only one variable defies this separation. That variable is internal task dependence, which appears in both the productivity models, but as a negative predictor. Interestingly, the perceived performance model is the domain where external and internal variables are both found to be significant predictors.

Overall, then, external variables are found to be important predictors of performance but not attitudinal

outcomes. Note that this conclusion is drawn upon analyses in which we seek to find the best predictors of an outcome from a set of design factors and group process variables. We are not saying that, when viewed in isolation, external variables do not predict attitudinal outcomes. For example, one can see from the correlation table that external communication is significantly correlated with job satisfaction. However, in the regression models, other variables proved more important than external communication in predicting satisfaction. Figure 3 summarizes the role of each design factor and group process variable in predicting work group outcomes.

5. Conclusion and future research

The results of this study indicate a separation between predictors of performance and attitudes for work groups in high-technology manufacturing settings. As hypothesized, external variables proved important predictors of a work group's performance, both actual (in terms of productivity) and perceived (as measured via self-report). However, external variables did not predict attitudes (in

8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1.00																		
0.13	1.00																	
-0.02	-0.12	1.00																
0.08	0.00	0.03	1.00															
-0.08	-0.33	0.03	0.02	1.00														
0.09	0.03	0.22	-0.07	0.01	1.00													
-0.16	0.02	0.39	-0.10	0.02	0.31	1.00												
-0.06	0.01	0.13	0.35	-0.02	-0.17	-0.09	1.00											
-0.14	-0.10	0.45	0.22	0.09	0.04	0.40	0.42	1.00										
0.13	0.12	0.08	0.08	-0.07	0.06	0.28	0.04	0.09	1.00									
0.22	0.23	0.07	0.21	-0.30	-0.07	-0.10	0.25	0.14	0.44	1.00								
0.01	-0.20	0.06	0.14	0.30	-0.09	0.02	0.29	0.26	0.33	0.17	1.00							
-0.19	-0.04	0.06	0.17	0.04	-0.34	-0.08	0.35	0.33	0.02	0.31	0.33	1.00						
-0.01	0.09	0.10	-0.10	-0.07	-0.06	0.01	0.01	0.16	-0.17	-0.11	-0.09	-0.12	1.00					
0.06	0.07	-0.14	-0.08	-0.01	0.03	-0.01	-0.02	0.01	-0.06	-0.26	-0.17	-0.09	0.12	1.00				
0.12	0.11	-0.15	0.02	0.01	0.00	-0.03	0.04	0.05	-0.03	-0.21	-0.12	-0.03	0.10	0.93	1.00			
-0.19	-0.13	0.26	0.29	0.10	0.12	0.09	0.46	0.37	0.20	0.13	0.25	0.28	-0.05	-0.11	-0.12	1.00		
-0.14	0.01	0.21	0.09	-0.20	0.00	0.22	0.10	0.12	0.05	0.16	-0.01	0.08	-0.08	-0.09	-0.09	0.11	1.00	
-0.27	-0.10	0.36	0.06	-0.12	0.08	0.46	0.42	0.50	0.16	0.13	0.11	0.23	0.17	-0.15	-0.17	0.34	0.37	1.00

Table 2. Regression models: a (–) indicates a negative beta value for this variable in the regression equation; all others are positive

<i>Model to predict</i>	<i>Adjusted R squared</i>	<i>F-value significance</i>	<i>Variables remaining in the regression equation</i>	<i>Cases</i>
Direct labor productivity	0.09	0.0073	external conflict	84
Indirect labor productivity	0.11	0.0107	internal task dependence (–) external conflict	84
Perceived performance	0.33	$<1 \times 10^{-4}$	external feedback (–) internal task dependence (–) external supportiveness external communication (–) external task dependence internal supportiveness internal conflict (–) monetary reward valence internal feedback	82
Extrinsic job satisfaction	0.08	0.0221	internal conflict (–) internal task dependence verbal reward valence	84
Intrinsic job satisfaction	0.55	$<1 \times 10^{-4}$	task cohesion (–) education (–) internal supportiveness internal communication internal conflict (–) task training verbal reward valence	84

this case, job satisfaction). Rather, internally oriented variables dominated the models for work group attitude.

To understand why external variables explained performance but not attitudes, we need to probe more deeply into the underlying group processes. Work group conflict provides a good arena for such an examination. Recent work on group conflict has separated the construct into two categories: relationship-oriented and task-oriented [17,18]. Relationship-oriented conflict centers on people and their relationships, and has been shown to be detrimental to work group effectiveness. Task-oriented conflict arises when there are fundamental disagreements regarding the group's task and its completion. Such conflict has been shown to be beneficial. Groups that engage in task-oriented conflict are considering different options and possibilities; while this process leads to conflict, it also enables groups to find the best solutions to problems. Only in excess is task-oriented conflict detrimental to group effectiveness.

In this paper, conflict was separated into external and internal components, not relationship and task-oriented ones. However, the qualitative data gathered in interviews and observation suggests that, in the setting of this study, external conflict was more likely to be task-oriented. Lack of proximity and infrequent contact are two key reasons why relationship conflict is less likely to occur with people external to the group. Engineers, for instance, have their offices outside the cleanroom facilities where production operators process wafers. Although some engineers make a point of donning a cleanroom suit at

least once every day to take a round of the cleanroom in the area whose processes they oversee, the norm is clearly to spend as little time as possible in the cleanroom. Technicians and supervisors spend much more time in the cleanroom, but as they support or supervise groups across the fab, they spend relatively little time in one place. Thus, production groups were typically in contact with engineers, technicians, and supervisors because a problem had arisen. This is not to say that relationships between work group members and outsiders did not form, but just that those relationships most often occurred in the context of a task-related dilemma.

As suggested earlier, high-technology workplaces by their very nature require increased monitoring and involvement on the part of technical knowledge workers such as engineers and technicians. One might reasonably expect that in such a work environment a work group in more frequent contact with external technical workers will have higher performance. That external conflict predicts performance seems to support this claim. Conflict may arise because a production problem has been unveiled. The work group and the engineering staff work together to find the cause and a solution. In the process, finger-pointing and accusations may lead to conflict, as may the imposition of a solution that is at odds with existing work methods. Overall, however, performance may be high because a problem has been uncovered and addressed. One should be careful to note that conflict between work groups and people external to them, such as engineers, in and of itself may not be beneficial. If not

Variable	Role in predicting work group outcomes
Task dependence	The more dependent group members are on one another in the completion of their tasks, the lower the group's actual performance, but the higher its extrinsic job satisfaction. The more dependent they are on people external to the group, the higher their perceived performance.
Training	Training proved largely non-predictive of both attitudes and performance. The sole appearance of a training variable is in the model for intrinsic job satisfaction, where task training is a positive predictor.
Autonomy	No autonomy variables appear in any of the models. Both administrative and technical forms of autonomy are positively correlated with perceived performance and intrinsic job satisfaction, but this association did not yield predictive value when combined with other variables.
Supportiveness	The supportiveness variables largely followed the external/internal dichotomy in terms of predictiveness, in that internal supportiveness appeared in the models for perceived performance and intrinsic job satisfaction, while external supportiveness appeared in the model for indirect labor productivity.
Rewards	Rewards appear in the models for satisfaction and perceived performance, but not for actual performance. Interestingly, verbal reward valence appears in both satisfaction models. One would have expected monetary reward valence to appear instead of verbal valence in the extrinsic model, but it does not. Rather, it appears in the model for perceived performance.
Feedback	External feedback is a negative predictor of indirect labor productivity; internal feedback is a positive predictor of perceived performance.
Communication	External communication proved a negative predictor of perceived performance while internal communication positively predicted intrinsic job satisfaction.
Conflict	An interesting difference between external and internal variables is seen with regard to conflict. External conflict is a positive predictor of productivity, while internal conflict is a negative predictor of perceived performance and satisfaction. Notably, each of the five models contains a conflict variable.
Cohesion	Cohesion variables are almost entirely missing from the regression models. Oddly, task cohesion is a negative predictor of intrinsic job satisfaction. Instead, one might have expected affective cohesion to be a positive predictor in this model.
Demographic variables	Education and organizational tenure were added as demographic variables. While tenure did not appear in any of the models, as expected, education was a negative predictor of intrinsic satisfaction. In other words, people with higher education were less satisfied with intrinsic elements of their work.

Fig. 3. Summary of the predictive role of design factors and group process variables.

accompanied with problem identification and resolution, it might in fact prove detrimental to effectiveness.

Internal conflict, by contrast, is as likely to be as much of a relationship nature as it is of a task nature. Production group members work together, often in close quarters, over the entire work shift. If an argument arises between two group members, they may not be able to put enough distance between themselves to allow for time to "cool off," especially in areas where they are dependent on one another to accomplish the task, as suggested by the significant positive correlation between internal task dependence and internal conflict. The factors of proximity, task dependence, and constant contact provide an environment in which internal conflict of a relationship nature might erupt. They also explain why such conflict might affect a worker's intrinsic job satisfaction. While external conflict might diffuse after the engineer or technician leaves the work area, internal conflict might fester as the two group members continue to work together over the course of the shift. The end result is dissatis-

faction with the work environment. On a positive note, the large negative correlation between internal conflict and team training, in which conflict resolution is typically a large part, suggests that internal conflict can be abated.

It seems fitting that the model for perceived performance is the one in which external and internal variables together prove predictive. Perceived performance was captured via self-report (like job satisfaction) to capture the work group's assessment of its own objective performance (such as productivity). Thus, it stands not as a measure of true performance, but rather captures something equally important: the group's sense of its effectiveness, which may or may not be in line with its true performance. (In this study, the measures of direct and indirect labor productivity are negatively correlated with perceived performance, but not significantly so.) The mixture of external and internal variables, coupled with the predictive capabilities of the former for objective performance and the latter for attitudes, suggests that work groups might be evaluating their performance in

terms of something other than quantifiable metrics. For example, a group might believe it is a high performer if its members get along well together, even if the quantity of wafers it produces each shift is lower than it should be. As a result, the perceived performance model may reflect the group's broadening of the meaning of performance to include team-related measures.

It should be noted that, although external variables predict performance outcomes better than internal ones do, they didn't explain a lot of the variance in performance. The model that explained the highest percentage of the variance is the one for intrinsic job satisfaction, at 55%. Other factors not captured in this study must be responsible for more of the variance in a high-technology work group's performance than the variables measured here. It is not clear whether those variables are more likely to be found among team design factors or group process variables. They could also be more related to larger environmental or organizational context factors which exist beyond the realm of the work group. Future research should explore what factors are most responsible for work group success in high-technology settings; unfortunately, follow-up data for the firms in this study have not been collected.

Implications of the current study are clear for managers in high-technology manufacturing firms: nurture relationships between technical staff and production work groups to boost actual performance, but expect satisfaction and other attitudinal measures to be most highly influenced by factors lying within the groups themselves. Resolve relationship-based conflict quickly, but worry less about external task-based conflict, especially if it is seen to be associated with problem identification and resolution.

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Biography

Diane Bailey is an Assistant Professor of Industrial Engineering and Engineering Management at Stanford University. Her research on work teams in semiconductor manufacturing won the 1995 Institute of Industrial Engineers Doctoral Dissertation Award and the 1998 *IEEE Transactions Engineering Management Society Publication Award*. Her current research focuses on the transformation of engineering work over time by technological and organizational factors. Her past work has appeared in engineering and business journals, including *Journal of Management*, *IEEE Transactions on Semiconductor Manufacturing*, *Organizational Dynamics*, *IEEE Transactions on Engineering Management*, and *Quality Management Journal*.

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